

Systematic Comparison of Hyperparameter Optimization Strategies for Winter's Classification of Impacted Third Molars Using EfficientNet-V2-S

Cherry B. Lisondra^{*a}, Jomari Q. Celis^a, Christopher R. T. Yap^a, Jose D. R. Boncales Jr.^a, Rogelio O. Badiang Jr.^a

^aMapúa Malayan Colleges Mindanao, Gen. Douglas MacArthur Highway, Matina, Davao City, Philippines 8000

ABSTRACT

Impacted third molars affect 16.7–68.6% of the global population, with Winter's Classification serving as the primary assessment approach for surgical planning. Deep learning has shown great potential for automating ITM classification, but current methods use ImageNet hyperparameters without testing them on the problems that dental radiography presents, such as class imbalance, small datasets, and safety requirements for clinical use. This study employs a two-phase methodology to compare hyperparameter optimization strategies, specifically baseline EfficientNet-V2-S, random search, and Bayesian optimization (Optuna). In Phase I, an unconstrained search space (learning rates: 10^{-6} to 10^{-1}) demonstrated that the baseline configuration surpassed both optimization strategies, as random search failed to converge (42.44% accuracy) due to an excessively high learning rate, whereas Bayesian optimization attained 81.44% in contrast to the baseline's 84.54%. These results indicate that automated optimization lacking appropriate search space constraints may yield inferior performance compared to well-established defaults. During Phase II, a refined search space (10^{-5} to 10^{-3}) allowed both approaches to modestly exceed the baseline, obtaining an accuracy of 86.77%, representing a 2.23% improvement over the baseline. Interestingly, both approaches independently arrived at a consensus configuration: batch size 16, focal loss gamma ≈ 2.0 , dropout 0.40, and a learning rate $\approx 3\text{--}5 \times 10^{-4}$, which empirically confirms these values as a robust hyperparameter setup for dental radiography. Although the enhancements in performance are slight, this research provides the first systematic evidence-based method for hyperparameter selection in Winter's Classification, proving that search space design is vital for successful optimization and offering reproducible guidance for future dental AI studies.

Keywords: Hyperparameter Optimization, Winter's Classification, Impacted Third Molars, Bayesian Optimization, Random Search, EfficientNet-V2-S, Dental Radiography, Deep Learning

1. INTRODUCTION

Impacted Third Molars (ITMs) are among the most common dental issues worldwide, with prevalence rate ranging from 16.7% to 68.6% across different populations¹. These teeth are unable to erupt properly due to obstruction or positional deviations^{2,3}, leading to complications including pericoronitis, cysts, and adjacent tooth damage⁴. Because of the complexity and the varied difficulty associated with ITMs extraction, accurate classification systems are necessary for surgical planning and formulation of an optimal treatment plan^{5,6}.

Among existing classification systems, Winter's Classification is a well-established and widely employed methodology for categorizing impacted third molars based on their anatomical angulation⁷. This system categorizes ITMs according to their angulation relative to the long axis of the second molar⁸ and is essential for treatment planning and reducing the risk of complications⁹. Winter classifies third molars as vertical, horizontal, distoangular, mesioangular, transversal, or inverse¹⁰. Classifying the impaction and level of difficulty in the third molar extraction is crucial for clinicians to determine the most suitable treatment and reduce potential adverse effects⁶.

Although deep learning techniques have been proposed for automating ITM classification, their implementation in Winter's system encounters specific obstacles that hinder the direct application of methods effective in general computer vision. Unlike ImageNet, which provides over 14 million labeled images spanning thousands of categories^{11,24}, dental

*cblisondra@mcm.edu.ph

imaging datasets are much smaller in comparison. A recent systematic review found only 16 accessible dental imaging datasets globally, comprising around 10,450 total images, which are significantly smaller than the available dataset in medical radiology¹². Aside from the lack of data, Winter's classification shows a severe class imbalance, which is common in the clinical field: certain angulations are much more common in patient presentations than others, such as transverse and inverted impactions, which make up less than 1% of cases¹³. This class imbalance makes traditional machine learning algorithms favor common classes while ignoring the rare but clinically important cases, potentially leading to misclassification which compromises patient health¹⁴.

Despite these documented challenges, most published dental AI studies use hyperparameters that were directly inherited from ImageNet pretraining or chosen through informal manual adjustment. Systematic reviews reveal inadequate methodology and incomplete reporting of hyperparameter selection procedures¹⁵. This approach assumes that hyperparameters that work well for classifying large, balanced natural images will also work well for small, severely imbalanced medical imaging tasks. This assumption has not been thoroughly tested¹⁶. More importantly, there are no studies that have systematically compared different hyperparameter optimization methods, like random search and Bayesian optimization, for Winter's classification or dental image analysis in general. This gap implies that practitioners don't have any evidence-based recommendation on how to properly tune deep learning models for dental diagnostics. This could lead to suboptimal model performance, which could affect clinical decision-making.

This research aims to address this gap by achieving three primary objectives: (1) systematically compare the default configurations of EfficientNet-V2-S, random search techniques, and Bayesian optimization (Optuna) through a two-phase method, aiming to understand the conditions that lead to the success or failure of each strategy; (2) evaluate whether both optimization techniques arrive at similar optimal hyperparameter configurations within a properly constrained search space, thereby validating their robustness and reproducibility for dental radiography; and (3) assess the performance of each approach across the two phases while identifying the clinical safety implication of the resulting configurations, aiming to offer evidence-based recommendations for hyperparameter selection in dental AI research. Although this research does not introduce a new architecture, its main contribution is the development of a systematic and reproducible approach for selecting hyperparameters in dental radiography. It offers the first evidence-based framework for Winter's Classification, which can be utilized and further refined by future researchers.

2. METHODOLOGY

2.1 Research Design

The study employs a comparative research design, as it allows the researchers to identify similarities and differences between multiple approaches in order to determine which model performs best under the defined circumstances and specified hyperparameters. This design ensures that the study provides a systematic and objective evaluation of each model by applying consistent datasets, preprocessing procedures, training conditions, and evaluation metrics, thereby enabling a fair and reliable comparison of their classification performance.

2.2 Research Locale

This study was conducted in collaboration with selected dental clinics in Davao City that offer panoramic X-ray imaging services and treat patients with impacted mandibular third molars. These institutions provided access to anonymized panoramic X-ray datasets and clinical expertise through labeling of the Winter's Classification of each panoramic X-ray for the study. The facilities were chosen based on their availability of high-quality imaging equipment and the presence of experienced dental professionals capable of validating the data and classifications.

2.3 Ethical Consideration

This study handles all the data acquired from partner dental clinics under strict confidentiality and ethical compliance. All files are anonymized before use via pseudonymization, with any re-identification key kept only at the clinic. Institutional ethics approval will be secured before data collection, and access will be limited to the research team. A formal data-sharing agreement will govern transfer and use, ensuring compliance with applicable privacy regulations such as HIPAA, GDPR, and the Philippine Data Privacy Act. These foster trust between the parties and establish clear accountability in the event of any breach.

2.4 Dataset and Preprocessing

Approximately 1,920 panoramic dental radiographs collected were retrospectively from three sources: the Tufts Dental Database, the Uribdodontics Dental Clinic, and the Alexian Brothers Health and Wellness Center. This preprocessing pipeline is utilized in conjunction with a related study investigating EfficientNet-V2-S across various multi-task learning configurations for the classification of affected mandibular third molars, as both studies employ the identical dataset. This study exclusively employs the Winter's Classification subset derived from that pipeline. All data were gathered with institutional ethics approval and rigorous confidentiality protocols, employing pseudonymization to anonymize patient identifiers. The study comprised adult patients aged 18 years or older, exhibiting clear visibility of third molars and adjacent structures.

Figure 1 depicts the exclusion criteria employed during data collection to ensure the excluded images don't negatively affect the model during training and to ensure dataset quality and clinical relevance. Exclusion criteria include patients below 18 years old since ITMs cannot be reliably evaluated prior to skeletal growth completion by age 17¹⁷, poor quality panoramic radiographs, and images without impacted third molars. This ensures that the model doesn't fail to perform in real-world scenarios.

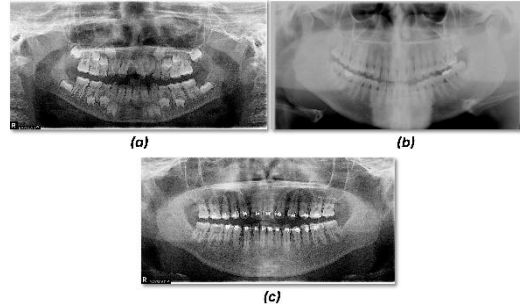


Figure 1. Exclusion criteria included (1) patients below 18 years of age, (2) inadequate visualization of molar regions, and (3) panoramic radiographs without impacted third molars.

Figure 2 illustrates the entire data preprocessing pipeline in this study. Because this study focuses on impacted mandibular third molars, only the lower half of the panoramic radiograph is utilized, specifically the third and fourth quadrants (Q3 and Q4). The upper portion of the PR was deemed unusable and therefore excluded during cropping.

We then transported the panoramic radiographs into the Label Studio, which is an open-source data labeling tool. Using this tool, 2 dentists with at least 2 years of experience in diagnosing and treating impacted third molars identified the presence of impacted third molars, specifically in areas under third and fourth quadrants (Q3 and Q4). If a specific section has an impacted molar, they simply annotate it as 'impacted,' and if there is none, as 'not impacted.' To ensure consistency and to lower the amount of variability between observers, a senior dentist verified the whole labeled dataset. Furthermore, custom annotations are also applied using Winter's and Pell & Gregory's classification systems, and the results are stored in CSV format. Since the Regions of Interest (ROI) are localized around the impacted teeth, utilizing the entire panoramic image is unnecessary for image classification¹⁸. Based on figure 2, we manually cropped two rectangles over two impacted teeth under third and fourth quadrants to extract the ROI, showing the apex of the mandibular third molar, the inferior alveolar canals, and the mandibular foramen and then rescale them to 244×244 pixels.

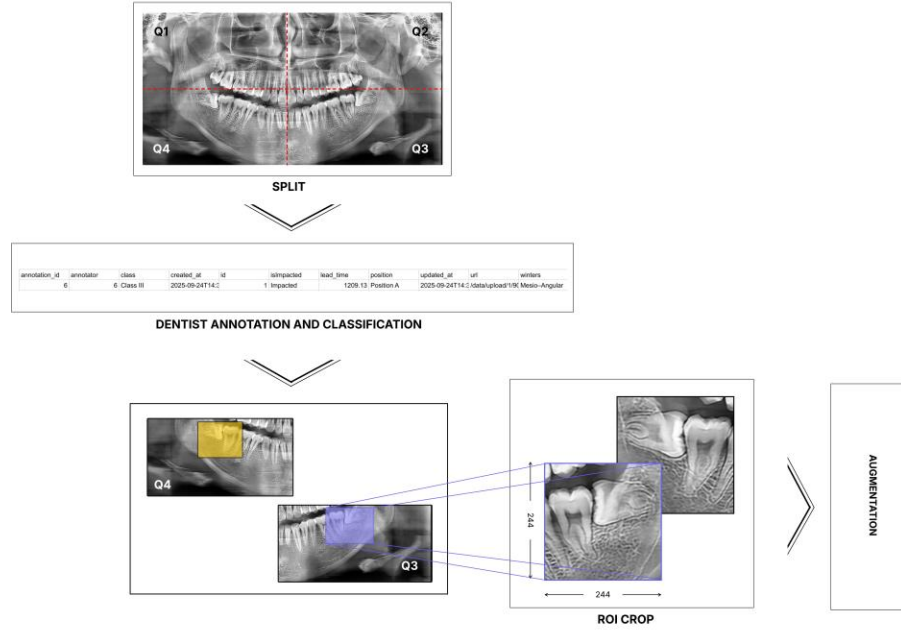


Figure 2. Data preprocessing pipeline. Panoramic radiographs (PR) are annotated using Label Studio for Winter's Classification and saved as CSV file. Impacted third molars are cropped from PR, rescaled to 244×244 pixels, and augmented to address class imbalance.

2.5 Dataset Augmentation

The final dataset of 1,920 images is subsequently augmented. As depicted in Fig. 2, the final stage of the pipeline involves augmentation which addresses class imbalance using five augmentation techniques. First is Random Resizing, which adjusts the dimension of an image¹⁹ to ensure consistent input size across the dataset. Second is Random Affine Transformations, which involve geometric changes that preserve the parallel lines in the image. These include translation, which moves the image left, right, up, or down to eliminate positional bias, and scaling, which adjusts the image to zoom in or out²⁰. Third is Color Jitter, which is the process of changing the red, green, and blue color channels to random values, resulting in a random change in the image's color²¹. Fourth is Random Horizontal Flip, which randomly flips the input image along its vertical axis with a specified probability²². Fifth is Gaussian Blur, which applies a Gaussian filter of varying kernel sizes to the original image²³, mimicking variations in image clarity that may occur across different imaging equipment. After augmentation, the dataset expanded to 3,000 panoramic images. By utilizing these data augmentation techniques, the model becomes more robust against class imbalance

2.6 System Architecture

The model classified mandibular third molars into the three most common types of Winter's Classification: mesioangular, vertical, and horizontal, allowing the assessment of predictive performance across clinically relevant categories. Input images were resized and normalized to standard dimensions to ensure consistent feature representation for the EfficientNetV2-S model. Various combinations and adjustments of hyperparameter settings were explored using Optuna hyperparameter tuning, aiming to conduct both wide and narrow search strategies to identify optimal configurations. Evaluation metrics such as accuracy, precision, recall, and F1-score were calculated for each class to provide a comprehensive and balanced assessment of the model's classification performance.

Figure 3 summarizes the study's full methodology pipeline. The pipeline starts with raw panoramic radiograph inputs, which are preprocessed before being sent into the EfficientNet-V2-S model for three-class Winter's classification. The anticipated outputs are then evaluated using performance measures, which include accuracy, precision, recall, and F1-score per class. Finally, a Naive Bayes analysis is performed on the three optimization strategies, namely baseline, random search, and Bayesian optimization, to discover the best model configuration.

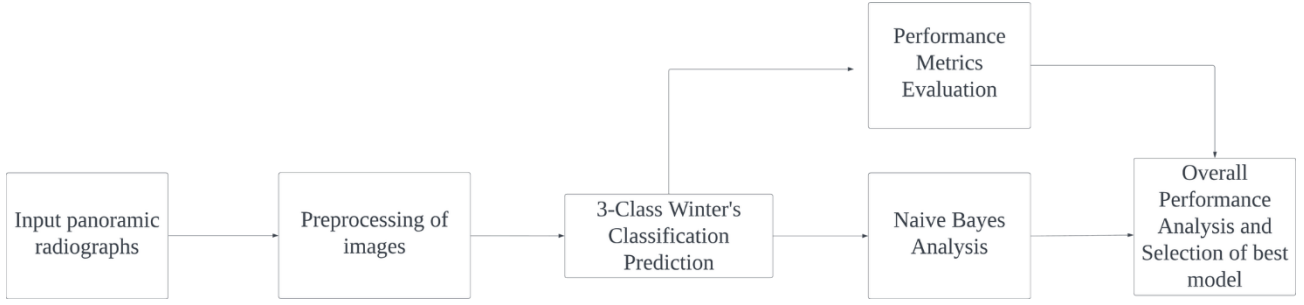


Figure 3. Overall methodology pipeline from input panoramic radiographs through preprocessing, 3-class Winter's Classification, performance evaluation, and Naive Bayes analysis for model selection.

2.7 Data Analysis

The data analysis evaluated the performance and interpretability of each hyperparameter optimization strategy in classifying Winter's impacted wisdom teeth. The analysis focused on four metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision}_{\text{macro}} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i} \quad (2)$$

$$\text{Recall}_{\text{macro}} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i} \quad (3)$$

$$\text{F1}_{\text{macro}} = \frac{1}{C} \sum_{i=1}^C \frac{2 * \text{Precision}_i * \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (4)$$

Where:

- TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives
- C = number of classes (Winter's: Mesioangular, Vertical, Horizontal)
- Macro averaging computes each metric per class then averages equally, which is appropriate for imbalanced datasets

3. RESULTS AND DISCUSSION

Table 1. Table Random Search and Optuna Comparison hyperparameter results Phase I

Hyperparameter	Baseline	Optuna	Random Search
Batch Size	16	16	16
Learning Rate	1e-3	4.37×10^{-3}	5.66×10^{-2}
Gamma	0.0	1.7	1.87
Dropout	0.2	0.27	0.40

Weight Decay	0	8.52×10^{-2}	4.57×10^{-3}
--------------	---	-----------------------	-----------------------

Table 2. Table Random Search and Optuna Comparison hyperparameter results “Narrow” Phase II

Hyperparameter	Baseline	Optuna	Random Search
Batch Size	16	16	16
Learning Rate	1e-3	2.83×10^{-4}	5.40×10^{-4}
Gamma	0.0	2.0	2.06
Dropout	0.2	0.40	0.41
Weight Decay	0	0.01	0.0063

Table 3. Efficiency analysis

Hyperparameter	Phase I: Exploratory	Phase II: Refinement Range
Learning Rate	$10^{-6} - 10^{-1}$	$10^{-5} - 10^{-3}$
Weight Decay	$10^{-6} - 10^{-1}$	$10^{-4} - 10^{-1}$
Focal Gamma	0.0 – 5.0	0.0 – 5.0
Dropout	0.1 – 0.6	0.2 – 0.5
Batch Size	16	16

Table 4. The Phase I “Wide” search area

Search Strategy	Best Accuracy			Macro F1			Macro Precision			Macro Recall		
Classification	Winters	Class	Position	Winters	Class	Position	Winters	Class	Position	Winters	Class	Position
Random Search	0.4244	0.6770	0.4003	0.2824	0.3678	0.2942	0.2738	0.4633	0.4081	0.3450	0.3733	0.3333
Bayesian (Optuna)	0.8144	0.6993	0.6649	0.8139	0.5396	0.6780	0.8254	0.5675	0.6727	0.8057	0.5292	0.6849
Baseline	0.8454	0.7337	0.7234	0.8427	0.5767	0.7280	0.8594	0.6064	0.7356	0.8318	0.5575	0.7261

Table 5. The "Phase II “Narrow” search area

Search Strategy	Best Accuracy			Macro F1			Macro Precision			Macro Recall		
Classification	Winters	Class	Position	Winters	Class	Position	Winters	Class	Position	Winters	Class	Position
Random Search	0.8677	0.7371	0.7251	0.8670	0.5781	0.7388	0.8677	0.6131	0.7437	0.8667	0.5557	0.7352
Bayesian (Optuna)	0.8677	0.7405	0.6993	0.8682	0.5630	0.7054	0.8754	0.6133	0.7184	0.8624	0.5352	0.6957

Baseline	0.8454	0.7337	0.7234	0.8427	0.5767	0.7280	0.8594	0.6064	0.7356	0.8318	0.5575	0.7261
----------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------	--------

To rigorously evaluate the hyperparameter landscape for Winter’s Classification, we conducted a two-phase optimization study comparing Stochastic (Random) Search, Bayesian Optimization (Optuna), and a deterministic Baseline (EfficientNetV2-S default settings).

In Phase I, the search space was unconstrained (Table 3), allowing learning rates to range from 10^{-6} to 10^{-1} and weight decay up to 0.1. This phase tested the algorithmic stability of the search strategies.

As shown in Table 4, the Random Search algorithm failed to converge, yielding a test accuracy of 42.44% for Winter's Classification. A detailed inspection of the hyperparameters (Table 1) reveals that Random Search selected an aggressively high Learning Rate 5.66×10^{-2} which likely caused catastrophic forgetting of the pre-trained ImageNet weights. This instability is further evidenced by the complete collapse of the Macro-Recall (0.3450), indicating the model failed to generalize to minority classes.

In contrast, the Bayesian Optimizer (Optuna) demonstrated superior stability within the same volatile search space. It identified a safer, albeit suboptimal, configuration 4.37×10^{-3} achieving an accuracy of 81.44%. However, in this exploratory phase, the Baseline model (Standard EfficientNet) outperformed both tuning strategies with an accuracy of 84.54%, suggesting that without search space constraints, automated tuning can struggle to surpass a well-established default configuration within a limited trial budget.

Leveraging the stability insights from Phase I, the search space was refined (Table 3) to focus on learning rates between 10^{-5} and 10^{-3} .

In this constrained environment, both search strategies converged significantly, finally surpassing the Baseline performance. As detailed in Table 5, both Random Search and Optuna achieved an identical top 1 accuracy of 86.77%, representing a +2.23% improvement over the Baseline (84.54%). While the accuracy scores were identical, the Bayesian optimized model exhibited superior clinical safety metrics. Optuna achieved a higher Macro-Precision (0.8754) compared to Random Search (0.8677) and the Baseline (0.8594).

Table 2 highlights the convergence of the optimal hyperparameters in Phase II. Both search strategies independently converged on a strikingly similar configuration:

Batch Size: 16

Focal Loss Gamma γ : 2.0 (Optuna: 2.0, Random: 2.06)

Dropout: 0.40

Learning Rate: 3.5×10^{-4}

This consensus between stochastic and Bayesian methods provides strong empirical validation that these values represent the intrinsic "sweet spot" for feature extraction in dental radiography using EfficientNetV2.

4. CONCLUSION

This research presented a systematic two-phase evaluation of hyperparameter optimization approaches for automating Winter's Classification in dental radiography utilizing EfficientNet-V2-S. The results reveal that in Phase I, the unrestricted search space showed that the baseline setup was more effective than both optimization strategies, with random search experiencing severe divergence due to an excessively high learning rate and Bayesian optimization achieving only 81.44% compared to the baseline's 84.54%. This outcome highlights an important point that automated optimization is not automatically better than established defaults, and the search space design is essential for successful optimization. In Phase II, with a more refined and constrained search space, both random search and Bayesian optimization exceeded the baseline, reaching a maximum accuracy of 86.77%, which translates to a modest yet steady enhancement of 2.23% over the baseline score of 84.54%. Although this advancement is incremental, the more

noteworthy finding is in the convergence behavior: both strategies independently reached a remarkably similar consensus on hyperparameters, specifically a batch size of 16, a focal loss gamma of ≈ 2.0 , a dropout rate of 0.40, and a learning rate $\approx 3-5 \times 10^{-4}$. This algorithmic triangulation strongly supports the notion that these parameters constitute a robust and reproducible hyperparameter setting for EfficientNet-V2-S in dental radiography. Additionally, the Bayesian-optimized model exhibited superior macro-precision (**0.8754**), which minimizes the likelihood of false positives and presents clinically safer configuration for automated surgical planning. The significance of this research lies in establishing a systematic and reproducible methodology for selecting hyperparameters in dental radiography, which is an area that has received little attention in previous studies. However, the study acknowledges certain limitations. The dataset is derived from a small number of dental clinics, and future investigations should examine the performance of the identified optimal configuration across various datasets and imaging conditions to better confirm its generalizability. In addition, calibration analysis or uncertainty estimation in future studies would offer more robust empirical backing for the clinical safety assertions linked to the Bayesian-optimized configuration.

REFERENCES

- [1] Fangfang, X., Yuxin, G., Ahmadi, S., Chunyan, W., Arian, P., Ming, Y., Junbo, T. and Sijia, N., "A prospective randomized study on the efficacy of real-time dynamic navigation in deep horizontal mandibular third molar extractions," *BMC Oral Health* 24(1), 1234 (2024).
- [2] Erçal, P., "Prevalence of impacted teeth and related pathologies: A retrospective radiographic study," *Essent Dent.* 2(3), 108-113 (2023).
- [3] Saisha, J., Kapeshi, C., Siame, A., Mutemwa, S., Kafumukache, E., Gupta, K., Ngoma, T., Mwembeshi, H. and Erzingastian, K., "Clinical presentation of impacted third molar tooth and its effect on the adjacent tooth, in Lusaka, Zambia," *Anatomy Journal of Africa* 12, 2296-2302 (2023).
- [4] Zhang, J., Zhang, K., Zhou, X., Ye, L., Liu, Y., Peng, Y. and Pan, J., "Full life cycle changes of low impacted mandibular third molar associated cystic lesions and adjacent tooth root resorption: a retrospective study," *BMC Oral Health* 24(1), 515 (2024).
- [5] Veerabhadrapa, S., Vengusamy, S., Padarha, S., Iyer, K. and Yadav, S., "Fully automated deep learning framework for detection and classification of impacted mandibular third molars in panoramic radiographs," *Journal of Oral Medicine and Oral Surgery* 31, (2025).
- [6] Juodzbalsys, G. and Daugela, P., "Mandibular third molar impaction: review of literature and a proposal of a classification," *J Oral Maxillofac Res.* 4(2), e1 (2013).
- [7] Balel, Y., "Bibliometric analysis of international publication trends in impacted third molar surgery research (2000-2020)," *Br J Oral Maxillofac Surg.* 59(10), 1220-1226 (2021).
- [8] Zirek, T., Öziç, M. Ü. and Tassoker, M., "AI-Driven localization of all impacted teeth and prediction of winter angulation for third molars on panoramic radiographs: Clinical user interface design," *Computers in Biology and Medicine* 178, 108755 (2024).
- [9] Albayati, M. T. and Bede, S. Y., "Reliability of two difficulty indexes in predicting the surgical extraction difficulty of impacted mandibular third molars," *Journal of Oral Medicine and Oral Surgery* 29(1), 5 (2023).
- [10] Miclotte, A., Grommen, B., Cadenas de Llano-Pérula, M., Verdonck, A., Jacobs, R. and Willems, G., "The effect of first and second premolar extractions on third molars: A retrospective longitudinal study," *J Dent.* 61, 55-66 (2017).
- [11] Kohli, M. D., Summers, R. M. and Geis, J. R., "Medical Image Data and Datasets in the Era of Machine Learning- Whitepaper from the 2016 C-MIMI Meeting Dataset Session," *J Digit Imaging* 30(4), 392-399 (2017).
- [12] Uribe, S. E., Issa, J., Sohrabniya, F., Denny, A., Kim, N. N., Dayo, A. F., Chaurasia, A., Sofi-Mahmudi, A., Büttner, M. and Schwendicke, F., "Publicly Available Dental Image Datasets for Artificial Intelligence," *J Dent Res.* 103(13), 1365-1374 (2024).
- [13] Zaman, M. U., Almutairi, N. S., Abdulrahman Alnashwan, M., Albogami, S. M., Alkhamash, N. M. and Alam, M. K., "Pattern of Mandibular Third Molar Impaction in Nonsyndromic 17760 Patients: A Retrospective Study among Saudi Population in Central Region, Saudi Arabia," *Biomed Res Int.* 2021, 1880750 (2021).
- [14] Salmi, M., Atif, D., Oliva, D., et al., "Handling imbalanced medical datasets: review of a decade of research," *Artif Intell Rev* 57, 273 (2024).

- [15] Ndiaye, A. D., Gasqui, M. A., Millioz, F., Perard, M., Leye Benoist, F. and Grosogeat, B., "Exploring the Methodological Approaches of Studies on Radiographic Databases Used in Cariology to Feed Artificial Intelligence: A Systematic Review," *Caries Res.* 58(3), 117-140 (2024).
- [16] Raghu, M., Zhang, C., Kleinberg, J. and Bengio, S., "Transfusion: Understanding transfer learning for medical imaging," *Advances in neural information processing systems* 32, (2019).
- [17] Kalaiselvan, S., Ganesh, S.K.N., Natesh, P., Moorthy, M.S., Niazi, T.M. and Babu, S.S., "Prevalence and pattern of impacted mandibular third molar: An institution-based retrospective study," *J Pharm Bioallied Sci.* 12(Suppl 1), S462-S467 (2020).
- [18] Milani, O.H., Atici, S.F., Allareddy, V., Ramachandran, V., Ansari, R., Cetin, A.E. and Elnagar, M.H., "A fully automated classification of third molar development stages using deep learning," *Sci Rep.* 14(1), 13082 (2024).
- [19] Gocer, E., "Medical image data augmentation: techniques, comparisons and interpretations," *Artif Intell Rev.* 56, 12171-12215 (2023).
- [20] Shorten, C. and Khoshgoftaar, T., "A survey on image data augmentation for deep learning," *Journal of Big Data* 6(1), 1-48 (2019).
- [21] Kumar, S., Asiamah, P., Jolaoso, O. and Esiowu, U., "Enhancing image classification with augmentation: Data augmentation techniques for improved image classification," *arXiv preprint arXiv:2502.18691* (2025).
- [22] Hao, R., Namdar, K., Liu, L., Haider, M.A. and Khalvati, F., "A comprehensive study of data augmentation strategies for prostate cancer detection in diffusion-weighted MRI using convolutional neural networks," *J Digit Imaging* 34(4), 862-876 (2021).
- [23] Ma, J., Hu, C., Zhou, P., Jin, F., Wang, X. and Huang, H., "Review of image augmentation used in deep learning-based material microscopic image segmentation," *Applied Sciences* 13(11), 6478 (2023).
- [24] Muñoz Guerrero, L. E., Ceballos, Y. F. and Trejos Rojas, L. D., "Leveraging ImageNet's Hierarchical Structure for Enhanced Image Classification and Retrieval," *Journal of Image and Graphics* **13**(4), 304-314 (2025).

AUTHORS' BACKGROUND

Your Name	Title*	Research Field	Personal website
CHERRY B. LISONDRA	Associate Professor	Machine Learning, Artificial Intelligence	
JOMARI Q. CELIS	Student	Machine Learning, Artificial Intelligence	
CHRISTOPHER ROSS T. YAP	Student	Machine Learning, Artificial Intelligence	
JOSE DONIE R. BONCALES	Student	Machine Learning, Artificial Intelligence	
ROGELIO O. BADIANG JR.	Associate Professor	Machine Learning, Artificial Intelligence	